BIO503 MID TERM REPEATED QUESTIONS

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1) **Order and disorder in biological system ?**

➢ Entropy is a measure of disorder: cells are NOT disordered and so have low entropy.
➢ The flow of energy maintains order and life. Entropy wins when organisms cease to take in energy and die.
➢ The Second Law of Thermodynamics states that the disorder, or entropy, in the universe is increasing. This also means that conversions of energy from one form to another are inefficient, with some energy being lost to cells as HEAT.

2) **What are factors that change entropy? (3 marks)**

Answer:

Factors can increase Entropy

1) Adding more particles increases the collisions and the randomness of motion

2) Increasing temperature The temperature can increase the velocity of particle motions 3) As volume increases the particles in a system allowed to roam in greater space; thus, more random motion 4) Increasing the freedom of motions of atoms and molecules - In general, the entropy of a system increases for processes in which: - Gases form from either solids or liquids - Liquids or solutions form from solids

3) **What is spontaneity? Describe the factors that contribute to spontaneity?**

Answer:

The tendency for a process to advance to equilibrium without external influence Something that happens naturally is spontaneous Some physical and chemical processes proceed naturally with no apparent cause or stimulus Any process will be spontaneous in one direction The reverse is non-spontaneous

• Change in enthalpy (ΔH) • Change in entropy (ΔS)

• Temperature of the reaction (T)

4) What are macroscopic and microscopic approaches of thermodynamics?

Behavior of matter can be studied by these two levels 1. Macroscopic approach—(Macro mean big or total)

2. Microscopic approach—(Micro means small)
Macroscopic approach: Certain quantity of matter is considered, without a concern on the events occurring at the molecular level. These effects can be perceived by human senses or measured by instruments. e.g: pressure, temperature Approach to thermodynamics is concerned with gross or overall behavior. This is known as classical thermodynamics.
Microscopic approach: The effect of molecular motion is Considered. e.g: at microscopic level the pressure of a gas is not constant, the temperature of a gas is a function of the velocity of molecules. A more elaborate approach, based on the average behavior of large groups of individual particles, is called statistical thermodynamics

dipole-dipole forces of attraction (Hydrogen Bondings) Two polar molecules Positively charged portion of one molecule is attracted to the negatively charged portion of another molecule

One molecule polar and other molecule not to be polar.

5) Gibbs free energy ?

Answer:

The energy associated with a chemical reaction that can be used to do work
➢ The Relation between Entropy ΔS and Enthalpy ΔH.

The equation is $\Delta G = \Delta H - T\Delta S$

6) Define membrane fluidity?

Answer:

Biological membranes are fluid dynamic that separate the inside from the outside of cells One remarkable feature of all biological membranes is their flexibility-their ability to change shape without losing their integrity and becoming leaky Lipid bilayer structure is quite stable.

Cell membrane: the thin and flexible boundary of a cell (also called a plasma membrane) It separates a cell from its surroundings. It allows nutrients to enter and waste to be removed.

The fluidity of the lipid bilayer is vital for the function of the membrane Membranes composed of phospholipids are highly mobile

Membrane fluidity is influenced by

Temperature

Composition

7) Define thermodynamics property? Explain intensive and extensive property?(5)

Answer:

A measurable quantity that defines the condition of a system e.g. Temperature T Mass P Pressure m Volume V Density Another useful property is specific volume defined as the volume of a unit mass

Types of Properties Thermodynamic properties are divided into two broad types:

- **Intensive properties**
- **Extensive properties**

Extensive properties

➢ Extensive properties are those whose values depend on the size-or extent-of the system.

• Mass m,

• Volume V,

• Total energy E These are examples of extensive properties.

➢ If mass is increased, the value of extensive property also increases

Intensive properties

➢ Intensive properties are those that are independent of the size of a system, such as

• temperature,

• pressure,

• Density

➢ Independent of mass

Extensive properties Extensive properties per unit mass are called specific properties. Some examples of specific properties are o specific volume (v = V/m), o specific total energy (e = Elm), and o specific internal energy (u = V/m). An example is specific volume, which has units of volume/mass, typically expressed as cubic feet per pound or cubic meters per kilogram

8) Application of biophysics at biological level?helix, beta-sheets in secondary structure of protein ?

Secondary structure: Alpha-helix The backbone follows a helical path (right, left), flexible Hydrogen bonds between backbone amino and carbonyl groups and those in the next turn of the helix The R-groups protrude out from the helix Proline tends to interrupt an alpha-helix

Secondary structure: Beta sheet Strands of protein lie adjacent to one another, interacting laterally via hydrogen bonds, between carbonyl oxygen and amino H atoms Successive side chains point straight up, then straight down.

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